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Batch No: 20RA08

Panel No:03

20RA274– Mini Project I

"Development OF LIBRARY AUDIT ROBOT USING BARCODE"

Type of Project: Application / Society Based

Expected Project Outcome : Product / Publication

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Review No : II

Date : 25.02.2026

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Introduction

- The existing library systems mainly rely on manual book arrangement and retrieval, while conventional robots are limited to barcode scanning and flat-surface movement, lacking the ability to climb ramps or handle uneven paths. To overcome this, the Smart Ramper Robot is designed to automate book delivery and return, assist elderly and differently-abled users, and support smart library management by reducing human effort, with future scope for inventory checking and real-time database updates.

- The work is carried out by designing a ramp-climbing mobile robot integrated with obstacle and slope detection sensors, controlled by a microcontroller for navigation and movement, and equipped with a simple book handling mechanism. The system is tested to ensure smooth ramp travel, safe handling of books, and efficient overall performance

Problem identification, scope and objectives (Contd.,)

Problem Definition

In libraries, arranging and collecting books is done manually, which is difficult for elderly and differently-abled users. Existing robots cannot move on ramps, so there is a need for a smart robot that can climb ramps, avoid obstacles, and handle books easily.

Scope

The Smart Ramper Robot is designed to automate book delivery and retrieval in libraries, with the ability to move on ramps and avoid obstacles. It reduces human effort, improves accessibility for elderly and differently-abled users, and supports smart library management. The project can also be extended for inventory checking, database updates, and integration with IoT or AI for advanced features.

Problem identification, scope and objectives

Objectives:

- To design and develop a robot capable of moving on ramps and flat surfaces in libraries.
- To automate book delivery, retrieval, and handling with reduced human effort.
- To assist elderly and differently-abled users by improving accessibility in library services.

Literature Survey – 01

Experiment	Inference	Problem gap
A mobile robot was designed to automate book searching and barcode scanning in a library. The system was capable of flat-surface navigation and book identification using line-following techniques.	The robot successfully reduced manual effort in book search and retrieval tasks,improving efficiency in library management. It proved the feasibility of integrating robotics into smart library systems.	Existing systems are limited to flat surfaces and cannot move on ramps or uneven paths. They also lack mechanisms to assist elderly or differently-abled users.

- **Source:**

A. Sharma, R. Gupta, Design of Autonomous Ramp-Climbing Service Robot for Indoor Applications, International Journal of Robotics and Automation, Vol. 39, Issue 2, 2024, pp. 112–120.

Literature Survey - 02

Experiment	Inference	Problem gap
The autonomous library audit robot was tested to scan barcodes of books on shelves and compare them with the library database. It successfully identified missing or misplaced books and updated the records in real-time with high accuracy.	The autonomous library audit robot using barcodes reduces manual effort and speeds up inventory checks with high accuracy. It ensures real-time updates, identifies misplaced books, and provides an efficient, cost-effective solution for large libraries.	Manual auditing of library books is slow, tiring, and prone to human errors. Conventional barcode scanning still requires human involvement for every shelf. Large libraries face difficulties in tracking misplaced or missing books quickly.

- **Source:**

New library Technologies and Trends in 2025 – notes use of autonomous robots for inventory audits.

Literature Survey - 03

Experiment	Inference	Problem gap
The autonomous library audit robot was tested in different aisles of the library to scan book barcodes and compare them with the master database. The robot navigated shelves at controlled speeds	From the experiment, it is inferred that the autonomous library audit robot using barcodes significantly improves the speed and accuracy of inventory checking compared to manual methods.	Traditional library auditing is slow, labor-intensive, and prone to human errors. Even with barcode systems, manual scanning requires staff to check each book individually.

- **Source:**

Toward Standardized and Technologically Enabled Library Inventory Management (2025) – highlights audit inefficiencies and need for digital audit tools

Literature Survey - 04

Experiment	Inference	Problem gap
The autonomous library audit robot was deployed to scan book barcodes across multiple shelves and compare the data with the library database. During the process, variations in speed, lighting, and barcode quality were tested to evaluate performance.	The study concludes that an autonomous library audit robot using barcodes offers a reliable, fast, and accurate alternative to manual auditing. It minimizes human effort, reduces errors,while effectively identifying missing or misplaced books.	Manual library audits are time-consuming, labor-intensive, and often prone to human error. Even with barcode systems, swhich is inefficient for large collections. Existing methods lack automation, and real-time data synchronization

Source:

Transforming Libraries. The Role of Autonomous Robots in Sorting, Inventorying and Book Search (2024)

Literature Survey - 05

Experiment	Inference	Problem gap
Deploy the barcode-based audit robot across three aisles with a known ground-truth catalog. Run at two speeds (slow/fast) under normal vs. low lighting while logging books/min and read accuracy. Seed shelves with controlled faults	The experiment shows that the barcode-based audit robot performs faster and more accurately than manual auditing. It successfully detected missing, misplaced, and unreadable books with minimal errors.	Large collections make it difficult to quickly identify misplaced or missing books. Current systems lack autonomous navigation and real-time database updating. Hence, there is a gap for a robotic system that can automate and speed up the audit process accurately.

- **Source:**

Transforming Libraries. The Role of Autonomous Robots in Sorting, Inventorying and Book Search" (2024)

Existing technology of the project

1. Barcode & RFID Systems:

Most libraries use barcode scanners and RFID tags for book identification and inventory. These systems are faster than manual entry and reduce errors. However, they still require staff to scan or track each book, making the process semi-automated.

2. Handheld Scanners & Smart Shelves:

Handheld devices and RFID-enabled smart shelves speed up the auditing process compared to manual checks. They allow staff to scan multiple books quickly and update records. Still, they depend heavily on human effort and cannot work independently.

3. Robotics & Vision-Based Systems:

Some research libraries and prototypes use robots with RFID/barcode readers and vision systems for autonomous auditing.

Source:

Transforming Libraries. The Role of Autonomous Robots in Sorting, Inventorying and Book Search (2024)

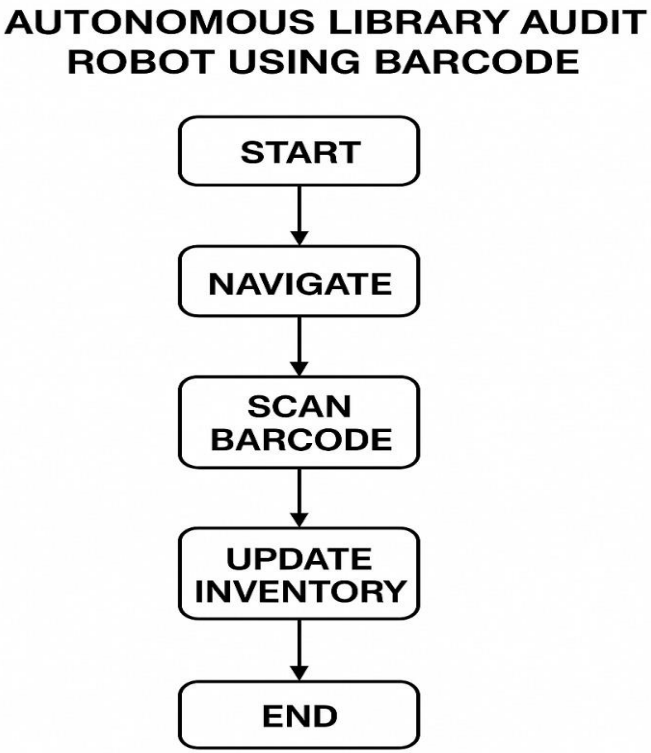
Existing technology of the project

Photographs:



Methodology to be adopted

Flow Chart



Timeline of the Project and Budget

Quarter Project Activity Timesheet

Chart Template with Months and Weeks



Design & Experimentation

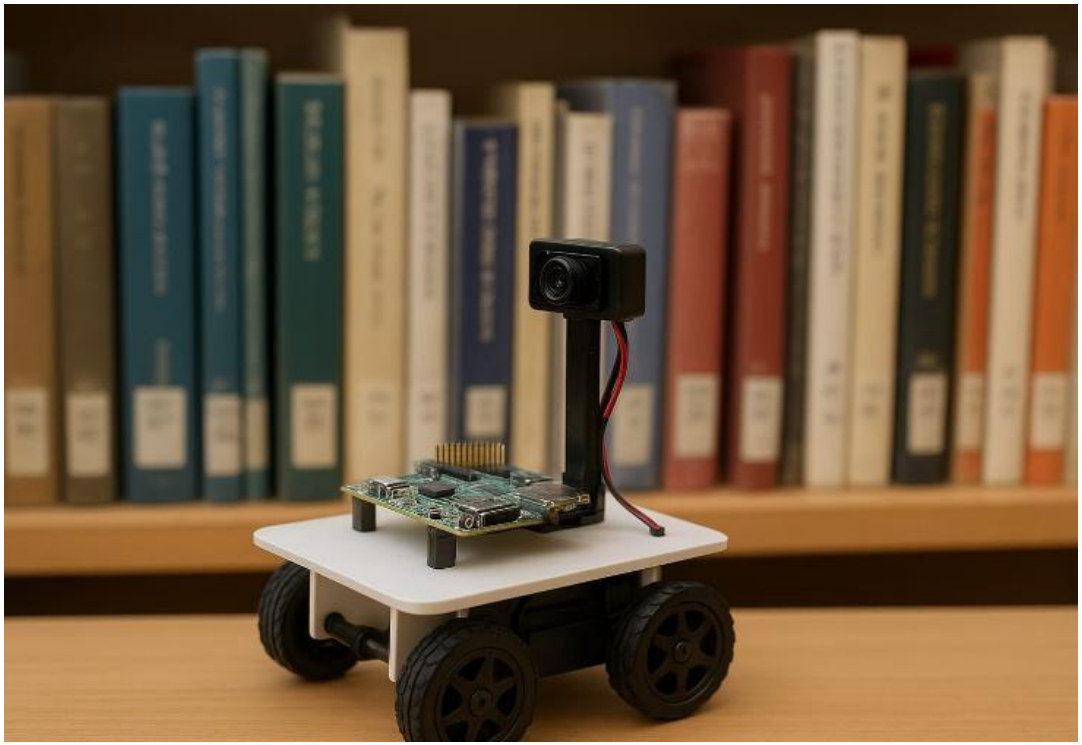
1. Base Structure:The robot is built on a lightweight acrylic base fitted with four wheels for smooth navigation through library aisles. The rectangular shape ensures proper balance and stability during movement.
2. Control Unit:A Raspberry Pi 4 board is used as the main controller to process inputs from sensors and the camera, manage barcode scanning, and control the robot's movement autonomously.
3. Vision System:A Pi Camera is mounted on an adjustable stand to scan barcodes placed on books across different shelf heights. The captured image data is processed by the Raspberry Pi for identification and record updating.

Design & Experimentation

4. Sensor and Motor Integration:Ultrasonic sensors detect obstacles while DC geared motors, controlled via a motor driver, provide precise movement. This combination enables the robot to move safely between shelves.

5. Material and Design:The chassis is made of lightweight acrylic or foam board to reduce load on motors. The modular design allows easy replacement of components like the camera or sensors for further testing and improvement.

Design & Experimentation



Results obtained till date

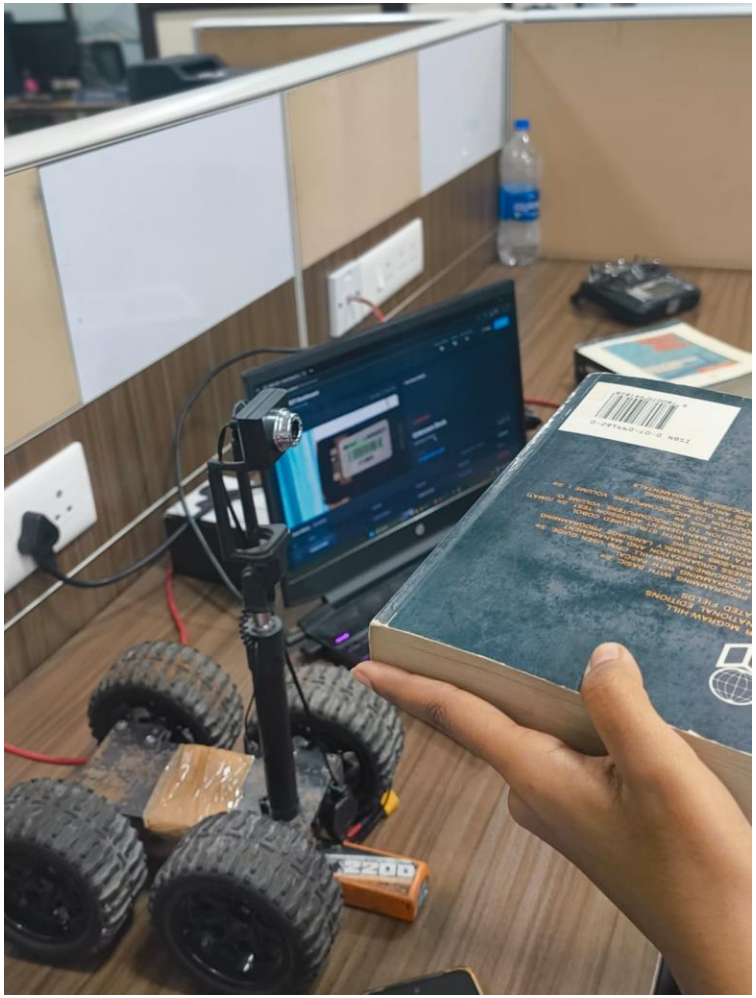
1. Raspberry Pi Integration:

The Raspberry Pi 4 has been successfully interfaced with the motor driver and power supply for basic control operations.

2. PiCam Testing:

The Pi Camera has been tested and verified for barcode detection and image capture functionality.3.

Results obtained till date



Works to be carried out

1. Hardware Setup:

The complete robot structure will be assembled, including the wheel base, motor driver, and power supply connection for stable operation.

2. Raspberry Pi Integration:

The Raspberry Pi will be connected with all sensors, motor driver, and power modules to act as the main control unit.

3. PiCam Connection and Testing:

The Pi Camera will be interfaced with the Raspberry Pi to capture and scan book barcodes placed on different shelf levels.

4. Barcode Scanning Program:

A Python-based program will be developed to detect, read, and decode barcodes from the captured images in real time.

5. Database Comparison:

The scanned barcode data will be compared with the existing library database to verify and update book information automatically.

Proposed Project Outcome

1.Autonomous Navigation : The robot will be capable of moving through library aisles automatically while avoiding obstacles using ultrasonic sensors.

2. Autonomous Book Scanning : The robot will move independently along library aisles and scan books without manual guidance.

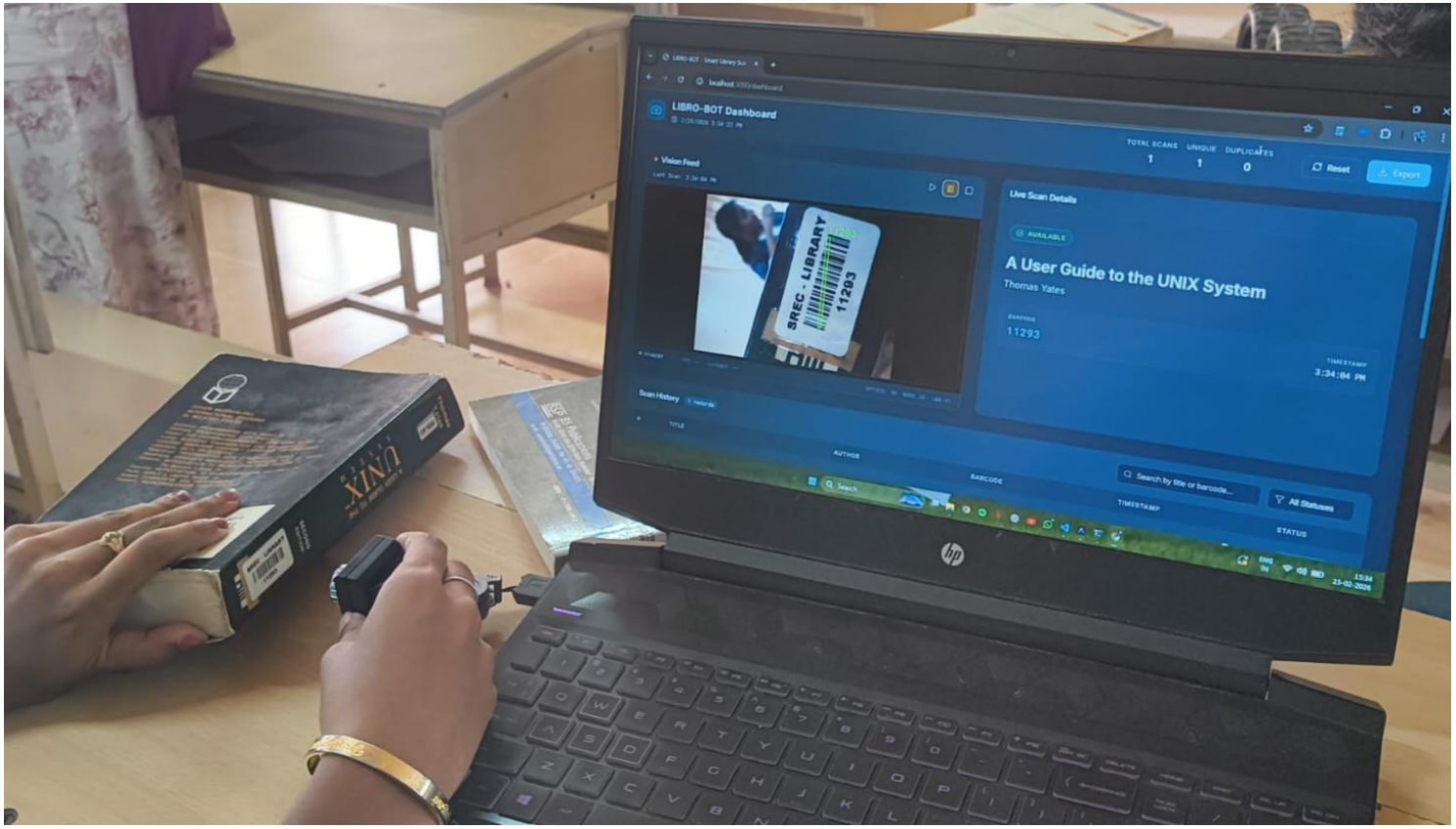
3. Accurate Barcode Reading : Using Pi Cam and Raspberry Pi, the robot will detect barcodes even on books placed at varying heights.

4. Real-Time Inventory Update : Scanned data will be instantly compared with the library database, highlighting missing or misplaced books.

5. Reduced manual Effort : Library staff workload will decrease as the robot performs repeated scanning and auditing tasks efficiently.

6. Expandable Design : The modular structure allows easy addition of new features, such as shelf mapping, RFID scanning, or voice alerts for missing books.

Photographs



Outcomes (Patent, Paper, Hackatons etc.) – applied / published / granted

1. Innovative Solution: Demonstrates a smart approach to library auditing using an autonomous robot with real-time barcode scanning.
2. Functional Prototype: A working robot capable of navigating aisles and scanning books using Raspberry Pi and PiCam integration.
3. Technical Excellence: Highlights skills in robotics, Python programming, sensor integration, and database management.
4. Efficiency & Accuracy: Provides faster and error-free library inventory checking compared to manual methods.
5. Hackathon / Publication Potential: Project has scope for participation in competitions, innovation showcases, and future research papers or patent applications.

References

1.https://www.researchgate.net/publication/381432646_Exploration_of_using_a_wall-climbing_robot_system_for_indoor_inspection_in_occupied_buildings

2.https://www.researchgate.net/publication/331587037_Automatic_Wall_Defect_Detection_Using_an_Autonomous_Robot_A_Focus_on_Data_Collection

3.https://www.researchgate.net/publication/363750602_Automated_wall-climbing_robot_for_concrete_construction_inspection

4. https://www.mdpi.com/2075-1702/13/6/521

5.https://www.sciencedirect.com/science/article/pii/S277299152200018